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# Taxon interactions control the distributions of cryoconite bacteria colonizing a High Arctic ice cap

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## Abstract

**Microbial colonization of glacial ice surfaces incurs feedbacks which affect the melting rate of the ice surface. Ecosystems formed as microbe–mineral aggregates termed cryoconite locally reduce ice surface albedo and represent foci of biodiversity and biogeochemical cycling. Consequently, greater understanding the ecological processes in the formation of functional cryoconite ecosystems upon glacier surfaces is sought. Here, we present the first bacterial biogeography of an ice cap, evaluating the respective roles of dispersal, environmental and biotic filtration occurring at local scales in the assembly of cryoconite microbiota. 16S rRNA gene amplicon semiconductor sequencing of cryoconite colonizing a Svalbard ice cap coupled with digital elevation modelling of physical parameters reveals the bacterial community is dominated by a ubiquitous core of generalist taxa, with evidence for a moderate pairwise distance–decay relationship. While geographic position and melt season duration are prominent among environmental predictors of community structure, the core population of taxa appears highly influential in structuring the bacterial community. Taxon co-occurrence network analysis reveals a highly modular community structured by positive interactions with bottleneck taxa, predominantly *Actinobacteria* affiliated to isolates from soil humus. In contrast, the filamentous cyanobacterial taxon (assigned to *Leptolyngbya/Phormidesmis pristleyi*) which dominates the community and binds together granular cryoconite are poorly connected to other taxa. While our study targeted one ice cap, the prominent role of generalist core taxa with close environmental relatives across the global cryosphere indicate discrete roles for cosmopolitan *Actinobacteria* and *Cyanobacteria* as respective keystone taxa and ecosystem engineers of cryoconite ecosystems colonizing ice caps.**

*Keywords*: bacteria, biogeography, cryoconite, ecosystem engineering, keystone species, Svalbard

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## Introduction

The interactions of glacial systems with climate, water and landscape assume considerable scientific and societal concern. This interest pre-dates the appreciation that glaciers, ice caps and ice sheets comprise microbial ecosystems (Hodson *et al.* 2008). Indeed, the activities of biodiverse microbial ecosystems associated with glacial habitats interact with both the dynamics of glacial systems and influence biogeochemical cycles (Edwards *et al.* 2014a; Hood *et al.* 2015; Rime *et al.* 2015). At the glacier surface, cryoconite ecosystems are recognized as

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major foci of microbial biodiversity and activity (Edwards *et al.* 2011; Cameron *et al.* 2012; Cook *et al.* 2015b) which influence ice surface albedo (Bøggild 1997; Takeuchi 2002) and potentially ice topography (Cook *et al.* 2015a).

The darkening action of granular microbe–mineral aggregates, termed cryoconite, upon ice surfaces causes localized melting and so cryoconite ecosystems often occupy quasi-circular holes within the ice surface which interact with the hydrology of the porous ice surface (Edwards *et al.* 2011; Cook *et al.* 2015c). The cryoconite biota includes viruses, bacteria, fungi, other microeukaryotes and meiofauna (Säwström *et al.* 2002) which actively contribute to carbon and nitrogen cycling (Hodson *et al.* 2007; Segawa *et al.* 2014). On both Arctic and Alpine glaciers, the composition and structure of cryoconite bacterial communities are closely related to the rates of microbial activities and the composition of cryoconite organic matter (Edwards *et al.* 2011, 2013a, 2014b). It appears that filamentous cyanobacteria (e.g. *Phormidium, Phormidesmis* or *Leptolyngbya* sp.) aggregate aeolian debris (Hodson *et al.* 2010; Langford *et al.* 2010), engineering the formation of granular cryoconite forming microbial communities distinctive from proximal habitats (Edwards *et al.* 2013b; Musilova *et al.* 2015). While the role of filamentous cyanobacteria is pivotal to the formation of stable cryoconite granules (Langford *et al.* 2010) harbouring a diverse community of bacterial heterotrophs, whether cyanobacteria represent keystone species or ecosystem engineers is equivocal (Edwards *et al.* 2014b). Similarly, while commonly occurring taxa in a given habitat, termed the core taxa, are assumed to regulate ecosystem functioning, and rare taxa (present within the long tail of a taxon abundance curve) may represent a store of genomic and functional variability as a 'seed bank' (Fuhrman 2009; whether core and tail taxa in cryoconite bacterial communities occur as generalists and specialists with broad- and narrow-shaped niches, respectively (Barberan *et al.* 2012; #833; Pedrós-Alió 2006), is poorly defined. Understanding the topology of the network of interactions between taxa varying in abundance and ubiquity (Steele *et al.* 2011; Barberan *et al.* 2012; Peura *et al.* 2015) can therefore be expected to enhance our understanding of how cryoconite bacterial communities colonize ice surfaces, accumulating organic matter and accelerating ice melt (Cook *et al.* 2015a,b).

Moreover, while previous studies have shown clear evidence of inter-regional and interglacier differences in cryoconite bacterial communities (Edwards *et al.* 2011; Cameron *et al.* 2012) the drivers and extent of spatial variation within the scale of individual glaciers are unclear. Recently, Langford *et al.* (2014) conducted a high-resolution sampling of cryoconite properties on a single Svalbard valley glacier, finding only moderate evidence for changes in the properties of cryoconite granules across the ice surface. Likewise Edwards *et al.* (2011) reported that interglacier differences outweighed very weak distance–decay relationships in bacterial community structure on three Svalbard valley glaciers. Furthermore, the temporal dynamics of cryoconite bacterial communities are less clear, with contrasting inferences made from intraseasonal sampling of cryoconite ecosystems at the margin of Greenland's ice sheet in two recent studies (Musilova *et al.* 2015; Stibal *et al.* 2015).

Consequently, the influence of seasonal melting upon community history or the response of cryoconite bacterial communities to environmental drivers prevailing within stable, low-gradient ice masses is unknown. This is likely to be the consequence of truncated environmental gradients associated with a low-complexity landscape responding to melt-associated drivers over a short dynamic range as the melting season proceeds rapidly. While studies of species turnover across elongated environmental gradients, for example at the ice sheet scale, could provide further insights, these will be across a broader, potentially continental, biogeographical scale as ice sheets span latitudinal and climatological gradients.

In contrast, ice caps provide an attractive model system for exploring the biogeography of microbial community development. Ice caps are defined as terrestrial ice masses which are not constrained by the topography of their underlying terrain but rather are shaped principally by their surface mass balance and, as distinct from ice sheets, have a surface area of less than 50 000 $km^2$ (Benn & Evans 2014). Consequently, by virtue of their surface topography, ice cap-associated microbial communities are likely to be situated within strong local environmental gradients within the same locality. Therefore, we hypothesize that the relative influence of dispersal, environmental and biotic filters in the assembly of cryoconite bacterial communities can be evaluated by, respectively, examining distance–decay relationships, linkages with physical parameters and taxon interactions of ice cap cryoconite microbiota.

In this study, we collected cryoconite from across an entire ice cap in the High Arctic archipelago of Svalbard which was constrained by a high-resolution digital elevation model, permitting a detailed analysis of the bacterial biogeography in relation to the topography of the ice cap. We show that geographic position and melt season duration do influence community structure, with evidence of a moderate distance–decay relationship in community similarity and the dominance of a core population comprising generalist taxa. Co-occurrence network analysis based identification of keystone species among heterotrophic bacteria rather than filamentous

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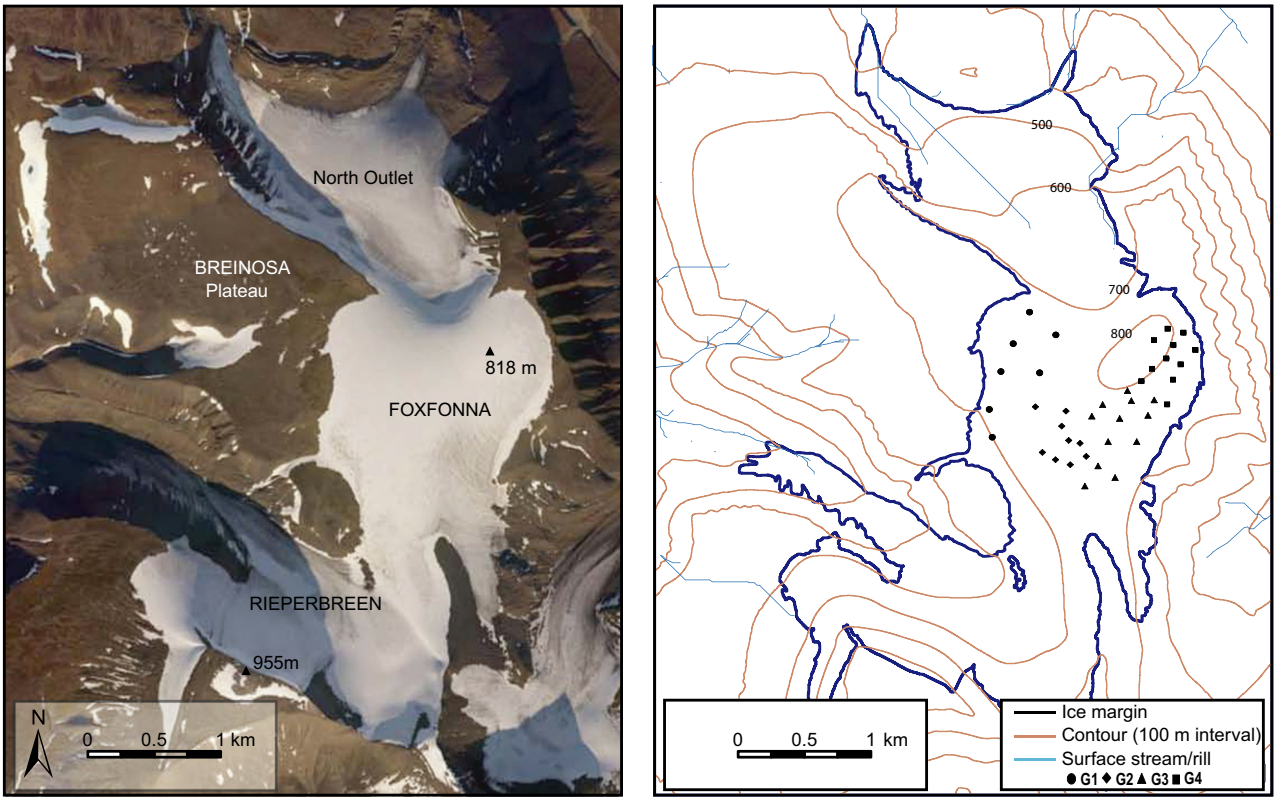


**Fig. 1** Map of study location, the Foxfonna ice cap on Svalbard with (A) aerial overview and (B) extracted digital elevation model, indicating sample points according to sector within the figure key.

cyanobacteria which are considered ecosystem engineers. We conclude that biotic filtering (i.e. taxon–taxon interactions such as competition or cooperation) plays a critical but hitherto unrecognized role in the microbial colonization of ice surfaces.

## Materials and methods

### *Site description and sampling*

Foxfonna is an ice cap measuring approximately 4 km$^2$ in central Svalbard (78° 08′N, 16° 07′ E; Fig. 1) with ice elevations ranging from ~675 to 955 m a.s.l (Rutter *et al.* 2011). The ice cap dome is almost decoupled from two small outlet glaciers descending to 285 m a.s.l.: Rieperbreen to the west and an unnamed outlet glacier to the north. Typically, the ice cap dome experiences melt for a short (~45 day) period of the summer. Surface mass balance measurements at seven stakes drilled into the ice cap indicate an average net annual balance of $-0.25 \pm 0.36$ (SD) m water equivalent for the period 2007–2014 (Rutter *et al.* (2011)). The strong variability is caused by occasional positive balance years for the ice cap, which last occurred during 2008 and 2012. Cryoconite samples were collected from the dome-shaped higher elevations (>700 m a.s.l.) of the ice cap on the 23rd of August 2011, towards the end of an ablation season during what was a close to average net mass balance year at the site (i.e. −0.38 m water equivalent). As is common at elevations proximate to the late summer snow line in Svalbard (Wadham *et al.* 2006), the superimposed ice layer was decaying and facilitated the development and exposure of cryoconite debris. Sampling was undertaken at four sectors according to aspect (hereafter G1, G2, G3, G4) over the ice cap surface. At 37 locations across the ice cap, cryoconite debris was aspirated into sterile 15-mL tubes and transferred on ice to −80 °C frozen storage within four hours for three weeks and thereafter transferred frozen in insulated containers within ten hours, to −80 °C storage in the UK. The surface area cover of cryoconite, termed, apparent cryoconite area (ACA) was calculated as previously detailed (Irvine-Fynn *et al.* 2010) while chlorophyll a was quantified from cryoconite slurries as described in Langford *et al.* (2014).

### *Digital elevation model*

Elevation data coupled with high-resolution aerial imagery were used to compile a digital elevation model

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(DEM) of the ice cap surface with a 5-m horizontal resolution. Due to the likely presence of noise in the raw elevation data, a standard smoothing filter was applied to the DEM (Wise 2000).

Primary (e.g. slope and aspect) and secondary (e.g. curvature and hydrological flow) indices were extracted from the smoothed DEM using ESRI's ArcGIS software. The indices describing the ice cap surface character and topographic attributes were retrieved using the ArcGIS 'Spatial Analyst' tool-set following established recommendations (e.g. Moore *et al.* 1991). While both slope and aspect dictate solar radiation receipt at the ice surface, slope also serves as a proxy for local meltwater flow velocity and an index for potential hydrological disturbance. Rather than using slope as a proxy for meltwater discharge, the flow accumulation area (hereafter, FAA) defined as the upslope area in $m^2$ draining to location point was used to represent a meltwater discharge regime. With knowledge that meltwater flow on Arctic glacier surfaces occurs dominantly through a near-surface perched aquifer (Irvine-Fynn *et al.* 2011; Irvine-Fynn & Edwards 2013), the topographic wetness index as a function of FAA and slope provides a continuous descriptor for areas over the ice cap likely considered to range between well drained or water saturated. The convex nature of the ice cap surface rendered use of the d8 algorithm (Jenson & Domingue 1988) more appropriate than alternatives (e.g. Tarboton (1997)) for prescribing flow routing over the ice cap surface.

Additional indices describing the surface conditions were calculated from the DEM. Potential incident radiation (IR) receipt for all locations across the ice cap throughout the summer melt season in 2011 was calculated following standard algorithms (Irvine-Fynn *et al.* 2014). Local variability in cloud cover precluded accurate, distributed estimations of actual radiation receipt at each sample site. However, spatially distributed air temperature records were extrapolated from data collected at the weather station on the outlet glacier to the north of Foxfonna (Fig. 1) using a local air temperature lapse rate of −0.65 °C per 100 m elevation. Measures of melt intensity in the form of a count of hours >0 °C (PositiveHrs), positive degree days (PDDs) and positive degree hours [PDHrs; see (Hock 2005)] were derived from the extrapolated weather station record.

This range of environmental parameters were extracted from the DEM for each sample site and normalized in Primer6/PERMANOVA+ (PRIMER-E Ltd) for use with multivariate analyses.

### *Sample handling and DNA extraction*

All samples were handled in a bleach-decontaminated laminar flow hood using sterile tools and certified DNA free plasticware as previously detailed (Edwards *et al.* 2011). Negative extraction and PCR controls were included to verify the absence of contamination based upon the absence of a band upon gel electrophoresis, but not sequenced. Community genomic DNA was extracted from 0.5 g of wet cryoconite using a CTAB/phenol–chloroform bead-beating-based extraction and polyethylene glycol precipitation (Griffiths *et al.* 2000) as previously described (Hill *et al.* 2015) and detailed in Data S1 (Supporting information). Reagents were DEPC-treated and autoclaved. DNA quality checks by agarose gel and preliminary 16S rRNA gene T-RFLP were performed as described previously (Edwards *et al.* 2014b) and extraction and negative template controls did not yield product.

### *16S ribosomal RNA gene amplicon semiconductor sequencing*

Bacterial 16S rRNA gene regions were PCR amplified using barcoded V1 and V3 primers (B-27F + MID; A1-357R; Table S1, Supporting information) in a single batch prior to semiconductor sequencing on a single Ion Torrent 316v2 chip exactly as described (Hill *et al.* 2015) and Data S1 (Supporting information). Amplification, library preparation and sequencing were conducted in a single batch. Sequence data are available at EBI-SRA (SRP067436: PRJNA306097).

### *Sequence processing and bioinformatics*

Resulting sequences were quality filtered in Python using Mothur (Schloss *et al.* 2009) with the USEARCH algorithm (Edgar 2010) before performing closed-reference OTU picking in QIIME 1.9.0 (Caporaso *et al.* 2010) using the Greengenes 13_8 reference database (DeSantis *et al.* 2006). OTUs were clustered at a threshold of 97%, and sequence taxa assignments and chimera checking were performed in QIIME using uclust (Edgar 2010) and RDP classifier version 2.2 (Wang *et al.* 2007). Permutational multivariate analysis of variance (PERMANOVA), canonical analysis of principal components (CAP) and distance-based linear modelling (distLM) were performed with fourth-root transforms of Bray–Curtis distances based upon OTU relative abundance, while the Mantel-based test RELATE was performed with 999 permutations using a resemblance matrix of the fourth-root transformed Bray–Curtis distances and pairwise physical distances. Default options were selected for CAP, including performance of leave-one-out analyses, an iterative cross-validation of model robustness. DistLM was performed using normalized predictor variables selected in stepwise protocol and their influence evaluated in sequential tests with

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adjusted $r^2$ values. PRIMER6/PERMANOVA + (PRIMER-E Ltd) was used for all multivariate analyses, and one-way ANOVA was calculated in Minitab 15. Data visualizations of OTU relative abundances using Microsoft Excel or PRIMER6/PERMANOVA and Adobe Illustrator are based upon unmodified data.

### *Network analysis*

A vector was created for each OTU to represent the OTU's abundance in each of the 37 samples as indicated by the formula:

$$x_i = [x_{i1}, x_{i2}, \ldots, x_{i37}](i = 1, \ldots, 755)$$

To reduce sequencing effort bias, $x_i$ values < 5 were set to zero (Zhang *et al.* 2013) and OTU vectors which contain less than 8 nonzero elements (20%) were removed to reduce false high correlations (Berry & Widder 2014). A second set of vectors was created based on environmental variables measured for each of the 37 samples. Pairwise Spearman correlations between all vectors were calculated and the associated p-value corrected for multiple comparisons with a Benjamini-Hochburg adjustment.

A community network was created based on significant correlations ($\rho > |0.7|$ and adjusted $P < 0.05$) using package [iGraph] in R (Csardi & Nepusz 2006), incorporating both OTU abundances and measures of environmental variables. Community detection was based on random walk algorithm ('walktrap') in [iGraph] (Pons & Latapy 2005). Network parameters were compared with the Erdös-Renyi random model of a network of equal size. For both observed and random model communities, network parameters were calculated using the [iGraph] package in R (Csardi & Nepusz 2006).

To identify keystone taxa, the community network structure was used to identify OTUs which function as 'bottlenecks' within the community, suggesting that they are central to community structuring and/or function. Bottlenecks are here defined as nodes with highest betweenness centrality, a count of the number times the bottleneck appears on the shortest paths between all other pairs of nodes (Peura *et al.* 2015) and therefore a measure of their connectivity within the co-occurrence network..

## Results

Semiconductor sequencing of 16S rRNA genes from 37 cryoconite samples distributed over the Foxfonna ice cap (Fig. 1) generated 4 609 547 total reads. Following processing, 755 bacterial OTUs were assigned by GreenGenes taxonomy to 13 phyla and 2 candidate phyla using a 97% similarity cut-off. It should be noted that results of preliminary T-RFLP community profiling of 16S rRNA genes cross-verified those of 16S sequencing in terms of spatial and environmental parameter prediction trends (data not shown); therefore, T-RFLP results are not reported further.

### *Community composition and relative abundance of higher grade taxa*

OTUs representing 87–91% of total relative abundance (RA) in the four sectors could be assigned to GreenGenes taxonomy (Fig. 2; Table S2, Supporting information). Across the cryoconite of Foxfonna ice cap the following phyla dominated the sequence data set: *Proteobacteria* (28.3% RA), followed by *Actinobacteria* (21.8% RA), *Cyanobacteria* (18.4% RA), *Bacteroidetes* (7.0% RA), *Chloroflexi* (5.5% RA), *Gemmatimonadetes* (4.85% RA) and *Acidobacteria* (1.77% RA). Within individual sectors (Fig. 2), *Proteobacteria* was the abundant phylum in sectors G1–G4 with RAs of 31.5%, 24.5%, 28.9%, 29.2%, respectively. *Actinobacteria* on the other hand were the second most abundant in G1 (23.9% RA), G2 (22.9% RA) and G4 (24.7% RA). The phylum *Cyanobacteria* was second most represented in the sequence data from sector G3 (21.7% RA) and third in sectors G1 (20.0% RA), G2 (14.6% RA) and G4 (17.5% RA). Within the *Proteobacteria*, *Betaproteobacteria* dominated over other classes (16.2% RA) followed by *Alphaproteobacteria* (6.8% RA). No significant differences were observed in the diversity indices for species richness (ANOVA, $F = 0.4$, $P = 0.754$) and evenness value (ANOVA, $F = 0.27$, $P = 0.845$) between sectors.

### *Evidence for a moderate pairwise distance–decay relationship in bacterial community structure*

Potential pairwise distance–decay relationships in Bray–Curtis similarity of fourth-root transformed OTU relative abundance and geographic distance were tested for with 666 pairwise combinations of holes at distances between 77 and 1664 metres. The distance–decay plot of the overall community (Fig. S1, Supporting information) shows a weak relationship between geographic distance (m) and community dissimilarity which is confirmed by RELATE analysis, revealing a moderate spatial influence upon overall community structure ($\rho = 0.275$, $P = 0.001$). This is accounted for by significant Spearman correlations (Table 1) for the *Acidobacteria* ($\rho = 0.256$, $P = 0.012$), *Chloroflexi* ($\rho = 0.36$, $P = 0.001$), *Gemmatimonadetes* ($\rho = 0.275$, $P = 0.025$), *Proteobacteria* ($\rho = 0.188$, $P = 0.045$) and unassigned taxa ($\rho = 0.397$, $P = 0.001$).

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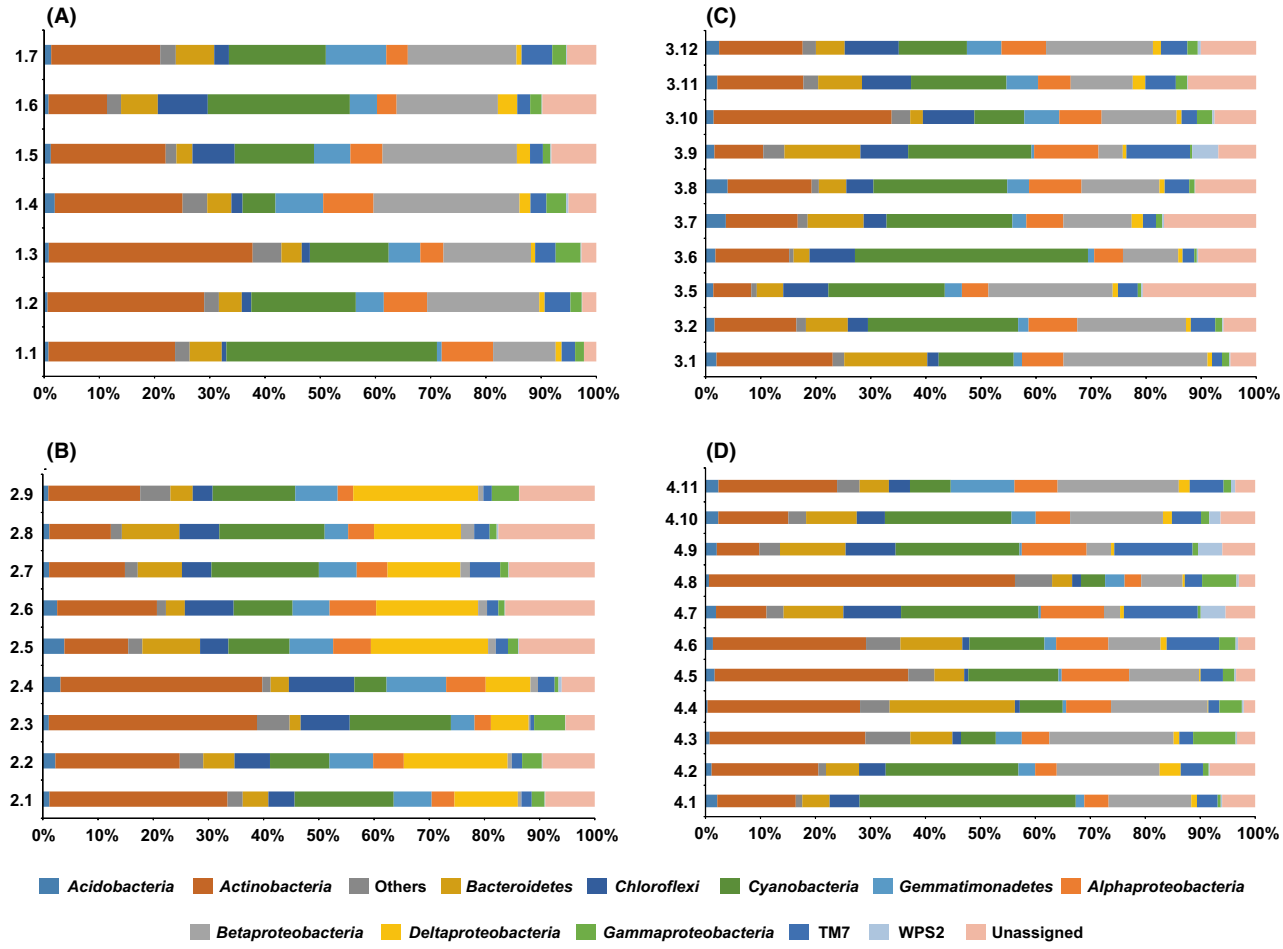


**Fig. 2** Phylum-level composition of 16S rRNA genes profiled by amplicon semiconductor sequencing and assigned to higher grade taxa according to the GreenGenes taxonomy (A) Sector G1; (B) Sector G2; (C) Sector G3; (D) Sector G4. The category 'Others' concatenates phyla present at a cumulative relative abundance <1% across the data set (Armatimonadetes, GN02, OD1, Thermi).

### *Environmental influences on bacterial community structure*

Canonical analysis of principal coordinates (Fig. 3B) clearly differentiates between sectors of the ice cap, assigning 78.3% of samples to the correct sector upon leave-one-out analysis. Moreover, PERMANOVA returns a highly significant sector effect (pseudo-$F$ = 3.0622, $P$ = 0.001 with 999 permutations). Pairwise PERMANOVA (Table S3, Supporting information) reveals each combination of sectors differ significantly ($t$ = 1.48–2.27, $P$ = 0.001–0.007) suggesting a clear effect of ice cap surface position on the bacterial community structure. When split by phylum, OTU relative abundances differed significantly for most phyla (Pseudo-$F$ = 2.43–5.74; $P$ = 0.001–0.007) between sectors with the exception of *Acidobacteria* and *Thermi*.

Therefore, to evaluate the relative importance of environmental factors in structuring the bacterial community, distance-based linear modelling (distLM; Fig. 3A) was

**Table 1** Spearman correlation between matrices of physical and Bray–Curtis distance by phylum provided via RELATE analysis; significant correlations highlighted

| Phylum | Rho | *P*-value |
|---|---|---|
| *Acidobacteria* | 0.256 | **0.012** |
| *Actinobacteria* | 0.123 | 0.114 |
| *Armatimonadetes* | −0.019 | 0.549 |
| *Bacteroidetes* | 0.113 | 0.141 |
| *Chloroflexi* | 0.36 | **0.001** |
| *Cyanobacteria* | 0.154 | 0.065 |
| *Gemmatimonadetes* | 0.219 | **0.025** |
| TM7 | 0.146 | 0.063 |
| *Proteobacteria* | 0.188 | **0.045** |
| *Thermi* | 0.094 | 0.174 |
| Unassigned | 0.397 | **0.001** |
| WPS-2 | 0.079 | 0.233 |
| OD1 | 0.086 | 0.203 |
| GN02 | −0.111 | 0.887 |

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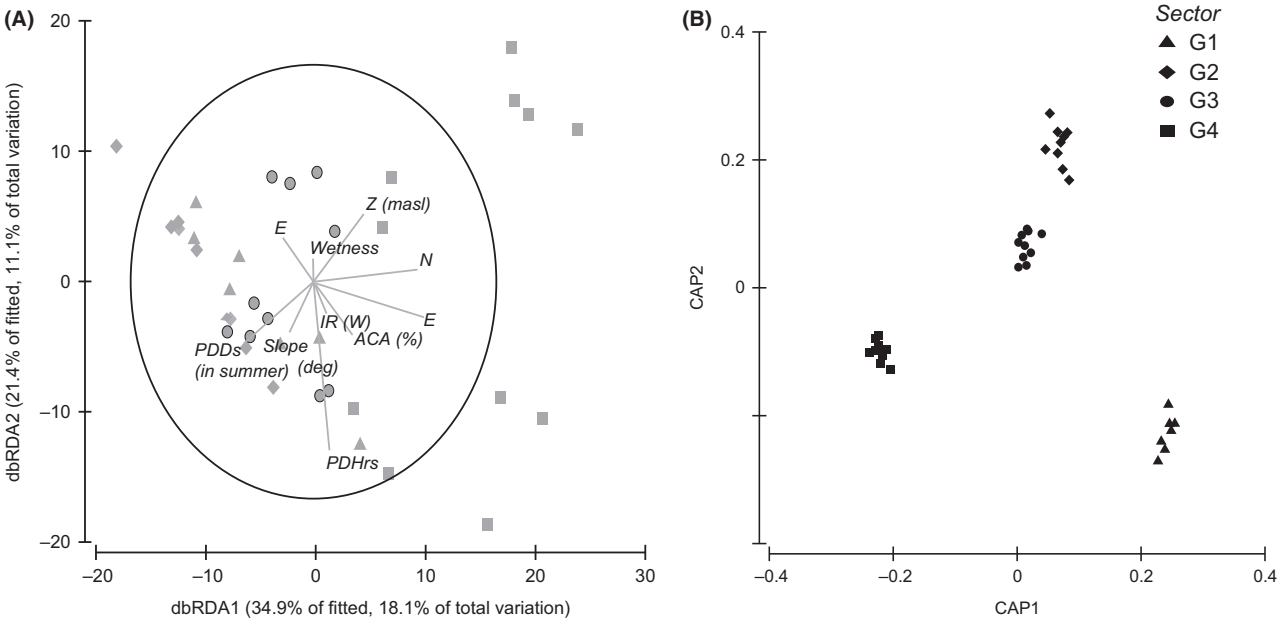


**Fig. 3** Ordination-based analyses of Foxfonna ice cap bacterial communities (A) distance-based redundancy analysis (dbRDA) ordination plot of distance-based linear models of physical parameter predictors of bacterial community structure; (B) canonical analysis of principal coordinates (CAP) of bacterial community structure according to a sector-based model.

performed, resulting in a model which explained 29.2% of the total variation in the first two distance-based redundancy analysis axes. Stepwise selection of predicting variables identified significant contributions ($P < 0.01$) by parameters related to geographic position (Northings, Eastings, elevation, slope and aspect), melting season duration (summer positive degree days and hours, positive hours, number of hours with incident radiation) and biotic factors (chlorophyll *a* concentration and apparent cryoconite area) in marginal tests (Table S4, Supporting information). Of these, sequential tests revealed positive degree days in summer as the most influential (contribution to adjusted $r^2 = 0.12$, pseudo-$F = 6.07$, $P = 0.001$) followed by slope (contribution to adjusted $r^2 = 0.08$, pseudo-$F = 4.93$, $P = 0.001$) Northings (contribution to adjusted $r^2 = 0.001$, pseudo-$F = 2.43$, $P = 0.001$), Apparent Cryoconite Area (contribution to adjusted $r^2 = 0.03$, pseudo-$F = 2.06$, $P = 0.01$) and Eastings (contribution to adjusted $r^2 = 0.04$, pseudo-$F = 2.99$, $P = 0.002$). Wetness, FAA, incident radiation, positive degree hours and elevation did not contribute significantly ($P > 0.05$) to the final model (adjusted $r^2 = 0.33$; $r^2 = 0.52$; Table S5, Supporting information).

### *OTU occupancy analysis reveals the cryoconite bacterial community is dominated by a generalist core*

To explore the distribution and dominance of specific bacterial taxa in cryoconite across the Foxfonna ice cap, the mean relative abundance of OTUs (clustered at 97%) across all samples was compared with the number of samples containing each OTU [i.e. occupancy; (Barberan *et al.* 2012)]. A clear pattern emerges (Fig. 4) in that the cryoconite bacterial community is strongly dominated by a small number of taxa. Of the 755 OTUs present in the data set, only 16 OTUs are present at a mean RA per sample >1%. The cross-sample cumulative RA of these 16 OTUs is strongly correlated with mean RA (Pearson $r = 0.99$, $P < 0.0001$) indicating minimal variation in their RA in sites across the ice cap. Strikingly, all 16 OTUs present at a mean RA per sample >1% are present in at least 36 of the 38 cryoconite samples analysed and indeed in all 37 samples for 13 of those OTUs. Consequently, these 16 OTUs are collectively considered a group of core taxa which is both ubiquitous and abundant within the cryoconite bacterial community. BLAST-based closest environmental relatives (CER) and closest named relatives (CNR) of core taxa (Table S6, Supporting information) reveal the core OTUs closely match uncultured sequences (CER %id 97–99%) mainly from cryospheric (13 OTUs) and soil (3 OTUs) habitats worldwide and more distantly related to cultivated taxa from soil (14 OTUs, CNR % id 88–98) plus Antarctic cyanobacteria (2 OTUs).

A long tail distribution of less abundant, variable occupancy, noncore taxa is also present (Fig. 4). Across both core and tail populations, mean RA is positively correlated with occupancy (Spearman $r = 0.77$, $P < 0.001$). The tail population of the cryoconite

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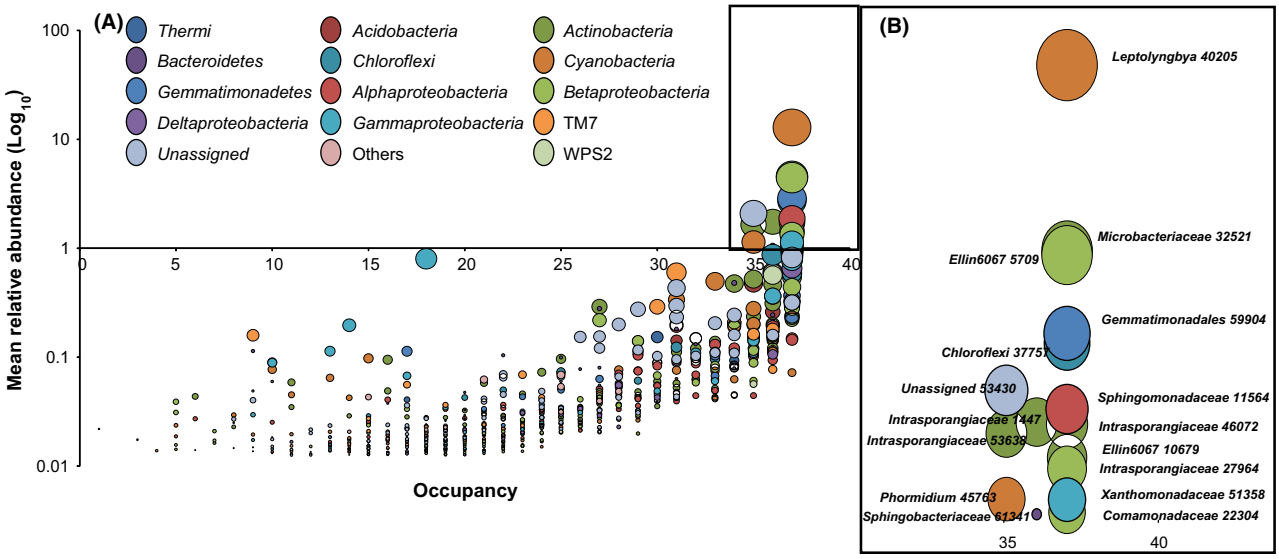


**Fig. 4** Occupancy plotting reveals that the Foxfonna ice cap cryoconite bacterial communities (A) are dominated by a core of generalist taxa highlighted by box (exploded view of core in inset B, annotated with OTU references). Bubble size is proportional to $\log_{10}$ of total RA, and bubbles are shaded by taxonomic affiliation. Occupancy is defined by the presence of an OTU within a site.

bacterial community on Foxfonna comprises OTUs affiliated to at least 9 phyla, including the proteobacterial classes *Alpha-*, *Beta-*, *Delta-* and *Gammaproteobacteria*. The core OTUs include representatives of *Actinobacteria* (5 OTUs), *Cyanobacteria* (2 OTUs), *Proteobacteria* (one *Alphaproteobacteria* OTU, three *Betaproteobacteria* OTUs and one *Gammaproteobacteria* OTU) and single OTUs from each of *Bacteroidetes*, *Chloroflexi*, *Gemmatimonadetes* and an unassigned OTU. Of these OTUs, an OTU, denovo40205, assigned to the filamentous cyanobacterial genus *Leptolyngbya* is very prominent, being present in all 37 sites and at a mean RA (12.5%) four times greater than the next most dominant OTUs, an actinobacterial taxon affiliated to *Microbacteriaceae* and a *Betaproteobacteria* OTU, both present at 4.4–4.5% mean RA. All remaining core OTUs are present at 1.0–2.9% mean RA and include an OTU affiliated to the filamentous cyanobacterial genus *Phormidium* (1.14% mean RA, at 36 sites).

### *Core OTUs are stronger influences on tail and total OTU relative abundances than environmental conditions*

The effect of core OTU composition and environmental parameters on tail and total OTU relative abundances was examined (Figs 5 and 6). Both core and tail OTU populations are significantly different in relative abundance across all quadrats of the Foxfonna ice cap (PERMANOVA; Pseudo-$F$ = 4.42, $P$ = 0.001; Pseudo-$F$ = 3.021, $P$ = 0.001, respectively; CAP shown in Fig. 6B, C). The Bray–Curtis distance matrices of core and tail OTU RA exhibit a much stronger correlation (RELATE; $\rho$ = 0.88, $P$ = 0.001) than to geographic distance (RELATE; $\rho$ = 0.29, $P$ = 0.01; $\rho$ = 0.27, $P$ = 0.001). Therefore, the interactions between core and tail OTU populations with environmental parameters were examined with a view to understanding the relative importance of core taxa and environmental conditions in shaping the cryoconite bacterial community.

When applying distLM with a matrix of core OTU RAs as predictors of tail OTU RA patterns (Fig. 6), all 16 core OTUs were very significant contributors ($P$ = 0.001–0.007) in marginal tests (Table S7, Supporting information), with 10 of 16 core OTUs highly significant in the derived model according to sequential tests (Table S8, Supporting information). For consistency, each core OTU is referred to by the most detailed taxonomic assignment made and the reference number of the OTU assigned during OTU selection. This model (adjusted $r^2$ = 0.57; $r^2$ = 0.76) is influenced most by an OTU assigned to *Sphingobacteriaceae* (hereafter referred to as *Sphingobacteriaceae*-61341; contribution to adjusted $r^2$ = 0.15, pseudo-$F$ = 7.43, $P$ = 0.001) followed by *Microbacteriaceae*-32521 (contribution to adjusted $r^2$ = 0.11, pseudo-$F$ = 6.18, $P$ = 0.001), *Intrasporangiaceae*-46072 (contribution to adjusted $r^2$ = 0.10, pseudo-$F$ = 5.27, $P$ = 0.001) with *Chloroflexi*-37757, *Intrasporangiaceae*-27964, *Gemmatimonadales*-59904, *Phormidium*-45763, *Leptolyngbya*-40205, *Xanthomonadaceae*-51358 and *Betaproteobacteria*-10679 in decreasing order of influence, yet remaining statistically significant (Table S8, Supporting information, $P$ = 0.001–0.025).

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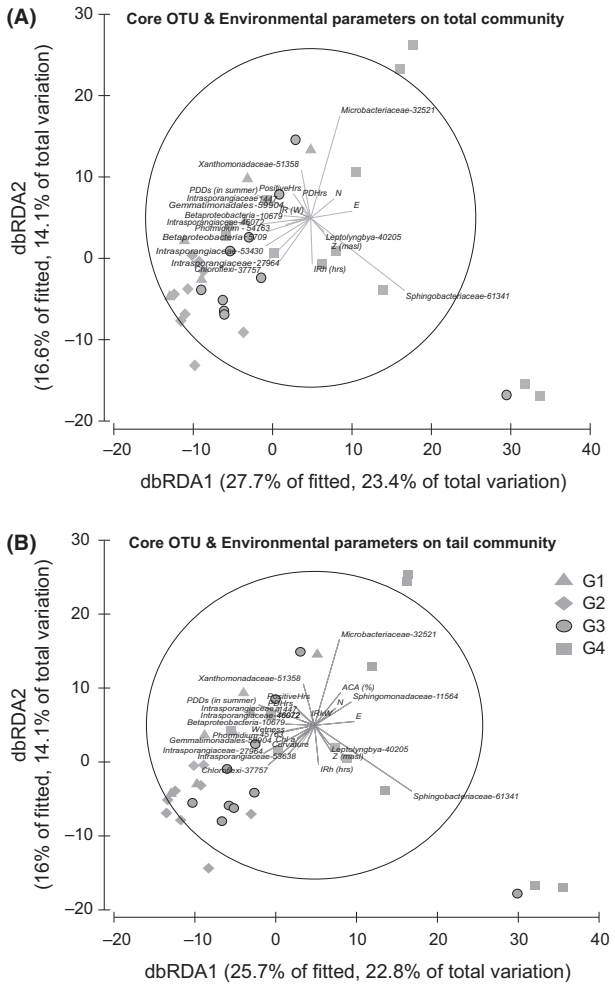


**Fig. 5** Distance-based redundancy analysis (dbRDA) ordination plot of distance-based linear models (A) core OTU and environmental parameters on total community and (B) core OTU and environmental parameters on tail community structure.

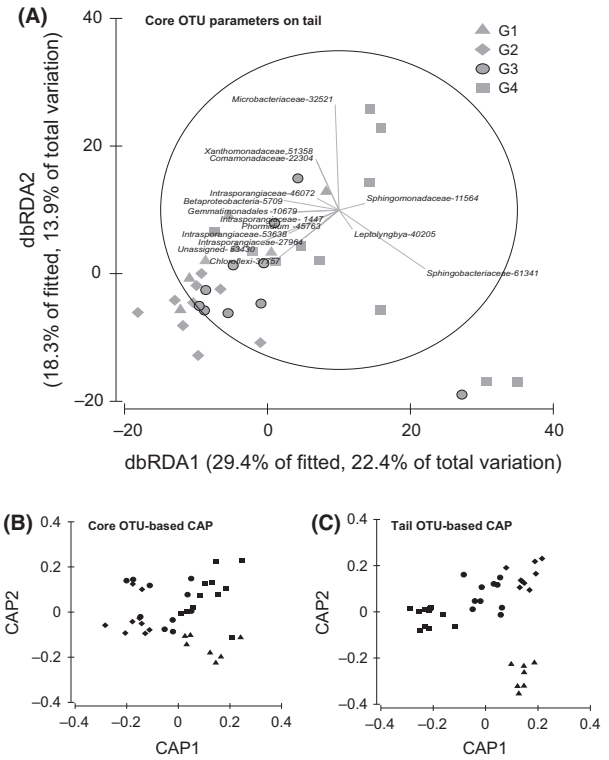


**Fig. 6** Distance-based redundancy analysis (dbRDA) ordination plot of distance-based linear models (A) of core predictors of tail community structure and canonical analysis of principal coordinates (CAP) of core and tail (panels B and C, respectively) community structures according to the sector.

Considering the apparent strength of core OTU influence in shaping the tail population, the relative influence of the core and environmental parameters upon the total and tail OTU populations was tested (Figs 5 and 6). All 16 OTUs and parameters relating to Cartesian position, chlorophyll content and apparent cryoconite area and melt season duration were significant predictors of total community structure ($P < 0.05$) in marginal tests (Table S9, Supporting information) while parameters relating to energy receipt and melt (e.g. incident radiation or wetness) were not, with the exception of hours of incident radiation. Sequential tests (Table S10, Supporting information) revealed the derived model (adjusted $r^2 = 0.60$; $r^2 = 0.84$) was heavily influenced by seven core OTUs (cumulative $r^2 = 0.59$), principally *Sphingobacteriaceae*-61341 (contribution to adjusted $r^2 = 0.17$, pseudo-$F = 8.55$, $P = 0.001$) and *Microbacteriaceae*-32521 (contribution to adjusted $r^2 = 0.13$, pseudo-$F = 7.35$, $P = 0.001$) followed by OTUs assigned to *Intrasporangiaceae*, *Leptolyngbya*, *Chloroflexi* and *Phormidium* in decreasing order of influence. Subsequently, the three least influential (but still statistically significant) predictors in the sequential tests were environmental parameters relating to geographic position and hours of incident radiation (their cumulative adjusted $r^2 = 0.04$, pseudo-$F = 1.52$–$1.97$, $P = 0.003$–$0.02$). The strong trend for core OTU influence to predominate over the environmental parameters measured in shaping the bacterial community is clearly paralleled in distLM prediction of tail population OTUs (Sequential tests: Table S11, Supporting information, dbRDA plot: Fig. 6, Marginal tests: Table S12, Supporting information) with *Sphingobacteriaceae*-61341 (contribution to adjusted $r^2 = 0.17$, pseudo-$F = 8.35$, $P = 0.001$) and *Microbacteriaceae*-32521 (contribution to adjusted $r^2 = 0.12$, pseudo-$F = 7.05$, $P = 0.001$) again the strongest predicting variables of tail OTU structure. A total of 8 core OTUs plus three environmental parameters (Cartesian position, hours of incident radiation) are significant predictors of tail OTU structures.

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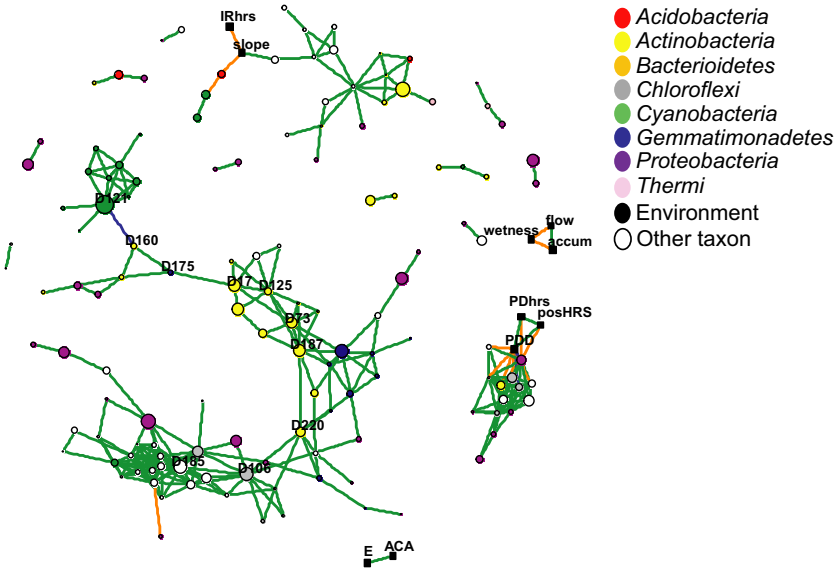


**Fig. 7** Community network based on significant pairwise Spearman correlations between OTUs (green – positive correlation, orange – negative correlation). Size of node is relative to average OTU abundance, while colour indicates OTU's phylum. Environmental variables have been included as nodes in the network and are indicated as black squares.

### *OTU co-occurrence network analysis reveals modular subnetworks*

Analysis of significant pairwise correlations between OTUs and environmental parameters resulted in a relatively small network with 145 nodes and 304 edges. The observed network was highly modular (observed = 0.77, Erdös-Renyi model = 0.41), with a considerably longer average path length than expected from a random model of the same size (observed = 4.93, Erdös-Renyi model = 3.58).

Environmental variables did not appear connected to most OTUs in the network, except for in the case of one small cluster of OTUs disconnected from the remaining network. This cluster was negatively correlated with several environmental variables related to energy inputs including positive degree day sum (PDD), hours with temperature above 0 °C (PositiveHrs) and the positive degree day hours (PDhrs) (Fig. 7).

The network contained several tightly clustered groups, disconnected or weakly connected with the remaining community (Fig. 7). Although there is some clustering of phylogenetic groups, most groups are made up of OTUs from diverse taxa. Only one group is clearly determined by phylogeny consisting of a small cluster of OTUs in the phylum *Cyanobacteria*.

Bottleneck OTUs as identified by the highest betweenness centrality score were dominated by OTUs of the phylum *Actinobacteria* (top ten bottleneck OTUs: Table 2). All bottleneck OTUs were connected to the largest cluster within the network through positive correlations, with the exception of *Leptolyngbya*-40205. This OTU links a tight cluster of *Cyanobacteria* OTUs through a negative correlation to OTUs in the largest network cluster (Fig. 7). Six of the ten top-scoring bottleneck OTUs are present within the core population (mean RA >1%).

## Discussion

### *The bacterial landscape of Foxfonna ice cap*

Understanding the distributions of microbiota provides insights into the assembly, biogeography and function of microbial communities across multiple scales (Bell *et al.* 2005; Bell 2010). In the context of cryoconite ecosystems, understanding the spatial organization of community composition provides insights into the microbial colonization of an extreme environment, and the consequential interactions with melt responses of glacial ice surfaces.

Here, we present the first bacterial biogeography of an ice cap. Semiconductor sequencing of bacterial 16S rRNA genes amplified from cryoconite ecosystems distributed across an Arctic ice cap reveals a cryoconite bacterial community dominated by a generalist core of (nearly) ubiquitous OTUs which influence the total and tail (i.e. noncore) bacterial community structure. Chief among the core OTUs is a taxon assigned to the filamentous cyanobacterial genus *Leptolyngbya* by the GreenGenes taxonomy. Co-occurrence network analyses reveal a highly modular network which is constrained by bottleneck OTUs (Peura *et al.* 2015) exhibiting high betweenness centrality scores. The ten top-scoring OTUs are also members of the core population. It is notable that the clearest phylogenetic signal within the network's modules is apparent in a module comprised solely of cyanobacterial OTUs

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**Table 2** Network bottlenecks identified as the OTUs with highest betweenness centrality metrics. Node ID matches that used in Fig. 7

| Node ID | OTU | Phylum | Class | Order | Family | Genus | Centrality | Mean RA>1 |
|---|---|---|---|---|---|---|---|---|
| D220 | Denovo61555 | *Actinobacteria* | *Acidimicrobiia* | *Acidimicrobiales* | C111 | | 1318 | |
| D106 | Denovo37757 | Chloroflexi | C0119 | | | | 1261 | + |
| D187 | Denovo53638 | *Actinobacteria* | *Actinobacteria* | *Actinomycetales* | *Intrasporangiaceae* | | 1102 | + |
| D17 | Denovo1447 | *Actinobacteria* | *Actinobacteria* | *Actinomycetales* | *Intrasporangiaceae* | | 973 | + |
| D175 | Denovo51679 | *Gemmatimonadetes* | *Gemmatimonadetes* | *Gemmatimonadales* | Ellin5301 | | 915 | |
| D160 | Denovo48894 | *Actinobacteria* | *Acidimicrobiia* | *Acidimicrobiales* | EB1017 | | 660 | + |
| D121 | Denovo40205 | *Cyanobacteria* | *Synechococcophycideae* | *Pseudanabaenales* | *Pseudanabaenaceae* | *Leptolyngbya* | 619 | + |
| D73 | Denovo27964 | *Actinobacteria* | *Actinobacteria* | *Actinomycetales* | *Intrasporangiaceae* | | 617 | + |
| D185 | Denovo53430 | | | | | | 552 | |
| D125 | Denovo41255 | *Actinobacteria* | *Actinobacteria* | *Actinomycetales* | *Intrasporangiaceae* | | 536 | |

Linking diversity analyses with geographic and other environmental parameters extracted from a digital elevation model of the ice cap permits the elucidation of the physical factors governing the assembly and structure of cryoconite bacterial communities. Distance-based linear modelling reveals geographic position on the ice cap and melt season duration to be better predictors of bacterial community structure than parameters relating to energy receipt or surface hydrology. However, the structure of the core OTU population is a much better predictor of both total and tail community structure than the measured environmental parameters alone. The stronger influence of specific taxa rather than physical conditions on the Foxfonna cryoconite microbiota is reflected within network analyses. Only two modules within the network are linked to the physical parameters, but by negative correlations. The analyses presented are highly coherent with the notion that cryoconite bacterial communities develop as a consequence of autogenic ecosystem engineering (Edwards *et al.* 2014b; Cook *et al.* 2015b) in the form of granular aggregation (Hodson *et al.* 2010; Langford *et al.* 2010). The ubiquity and dominant abundance of OTU *Leptolyngbya*-40205 within the 16S sequencing data are particularly consistent with a role as an ecosystem engineer (Musilova *et al.* 2015).

However, the prominence of heterotrophic bacteria within the core and bottleneck OTU populations is intriguing. In particular, six OTUs assigned to the *Actinobacteria* accounting for six of the top-scoring bottleneck OTUs and five of those are present within the core population. Of these, five are members of *Intrasporangiaceae.* While *Actinobacteria* have been detected in previous studies of cryoconite bacteria (Edwards *et al.* 2011, 2014b), this is the first time their prominent role in structuring the cryoconite bacterial community has been invoked. Previous work has identified the predominance of *Alpha-* and *Betaproteobacteria* within cryoconite microbiota (Edwards *et al.* 2014b; Stibal *et al.* 2015). While these are well represented within the core population of Foxfonna cryoconite, they are conspicuously absent from the bottleneck OTUs. We therefore infer that while taxa from *Cyanobacteria* engineer the ecosystem and *Proteobacteria* contribute to heterotrophic processes, certain *Actinobacteria* play a contrasting role by mediating key processes or biotic interactions which affect overall community structure. As keystone taxa are defined as taxa which show influence upon a community or ecosystem beyond that expected from their abundance (Power & Mills 1995), we consider these *Actinobacteria* OTUs as keystone taxa and that biotic factors may play a hitherto unrecognized role in the formation of cryoconite bacterial communities. Consequently,

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evaluating the relative roles of dispersal, environmental and biotic filters in shaping the cryoconite bacterial community is merited.

### *How does dispersal filtering shape this bacterial community?*

Contemporary microbial ecology literature is replete with studies inspired by Baas-Becking's infamous statement (Baas-Becking 1934; De Wit & Bouvier 2006). A broad consensus may be that some taxa are indeed cosmopolitan, while others exhibit biogeographical trends (van der Gast 2015). For glaciers, the predominance of cosmopolitan taxa has been noted (Darcy *et al.* 2011; Franzetti *et al.* 2013) and previous work indicated distance–decay effects had negligible influence in shaping the cryoconite biota of neighbouring valley glaciers (Edwards *et al.* 2011). As the Foxfonna ice cap is dome-shaped and not constrained by its surrounding topography to face a given aspect, unlike valley glaciers, we hypothesized that potential distance–decay effects would be revealed in this setting. Overall, there are statistically significant but moderate distance–decay effects, which consistent for both core and tail populations, and are pronounced and significant for *Chloroflexi*, *Acidobacteria* and *Gemmatimonadetes* in clear contrast to other phyla. It may be that specific traits in the life history of these taxa condition their dispersal (e.g. Chu *et al.* 2011; DeBruyn *et al.* 2011).

Meanwhile, it is noteworthy that all core OTUs have closest environmental relatives exhibiting >97% identity along the V1–V3 region of the 16S rRNA gene present in samples from a global range of habitats which, with one exception, are from the cryosphere (Table S6, Supporting information). Coupled with their ubiquity across the ice cap, most probably due to redistribution across the ice surface, this suggests an important trait of these core taxa is their broad distribution across the cryosphere, thus promoting their likelihood of colonization, resulting in a locally abundant and ubiquitous core population derived from a global pool of propagules.

### *How does environmental filtering influence this bacterial community?*

The second clause of Baas-Becking's statement (Baas-Becking 1934) directs the reader's attention to the notion that prevailing environmental conditions influence microbial community composition. Countless studies certainly support the importance of abiotic factors in shaping microbial communities influenced by deterministic processes [e.g. (Wood *et al.* 2008)]. The influence of surface hydrology and distance from the ice margin has been inferred for valley glacier and ice sheet microbiota, respectively (Edwards *et al.* 2011; Stibal *et al.* 2015). Here, distLM consistently invoked physical parameters relating to geographic position and melting season duration (either as positive degree days, Table S5, Supporting information; or hours of incident radiation, Table S11, Supporting information) as the strongest significant predictors of community structures.

A common feature of all models was that factors relating to surface hydrology, either as wetness of the ice surface or the extent of the FAA, failed to predict community structure. Within the context of the Foxfonna ice cap, which resides at higher elevation and latitude, with low mass balance gradients that exhibit strong interannual variability, this is plausible. These conditions mean that community development may be curtailed to relatively brief seasons of predominantly bare ice with a limited evolution of surface hydrological networks and porous weathering crust ice at the surface, in contrast to cryoconite situated on strongly ablating ice with longer growing seasons (Irvine-Fynn & Edwards 2013; Cook *et al.* 2015c). While temporal analyses of cryoconite microbiota at the margin of the Greenland ice sheet imply overall stability in community structure within melting seasons (Musilova *et al.* 2015), we infer that the overall duration of melting season is an influential environmental parameter. Further studies should directly examine the temporal evolution of cryoconite community structures, particularly in conditions beyond those typical of the south-western margin of Greenland's ice sheet, and in so doing challenge the potential oversimplification of current global models of cryoconite carbon cycling, which assume uniform rates of productivity across a melting season of fixed duration (e.g. 70 days: Anesio *et al.* (2009)).

### *How does biotic filtering in the form of taxon interactions influence the bacterial community?*

Recognition that biotic filtering (i.e. taxon–taxon interactions such as competition, cooperation and indeed ecosystem engineering) is an influential driver in the assembly of environmental microbial communities is much more recent (e.g. Goberna *et al.* (2014)). In the context of glacial ecosystems, discourse regarding biotic filtering has been limited to identifying algal taxa as primary colonizers of glacial surfaces (e.g. Lutz *et al.* (2015)) or the role of filamentous cyanobacteria as putative ecosystem engineers or keystone species (Edwards *et al.* 2014b). Here, we deduce that taxon interactions drive the assembly of cryoconite communities colonizing the Foxfonna ice cap. The composition of the core bacterial taxa is a strong predictor of the total and tail

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bacterial populations, with the distributions of specific taxa proving better predictors than any physical parameters. Moreover, the highly modular co-occurrence network is structured by nodes acting as bottleneck OTUs (Table 2).

Congruent with prior work (Hodson *et al.* 2010; Langford *et al.* 2010), we find filamentous cyanobacteria; specifically, the OTU *Leptolyngbya*-40205 are important in the cryoconite bacterial community, but while we concur they represent autogenic ecosystem engineers (West 1990; Langford *et al.* 2010; Edwards *et al.* 2014b), they are less prominent as keystone taxa. In contrast, selected *Actinobacteria* OTUs are prominent, highly centralized bottleneck OTUs within the highly modular co-occurrence network observed (Fig. 6). We infer these represent keystone taxa in that their influence exceeds their relative abundance within the community profiled. Their positive co-occurrence with other taxa mitigates against their role in competitive exclusion. As their closest named relatives (Table S6, Supporting information) comprise taxa associated with soil humus, we speculate these taxa may play roles in the humification of dark organic matter associated with cryoconite (Takeuchi *et al.* 2001).

Others have interpreted phylogenetic clustering as evidence for biotic filtering (Goberna *et al.* 2014). The co-occurrence network derived here is modular in its nature, with taxonomically diverse modules, a characteristic in common with taxon co-occurrence networks established for other polar habitats, and interpreted as a sign of metabolic plasticity (Vick-Majors *et al.* 2014). Thus, the predominance of phylogenetically diverse modules may permit the functional stability of the community in the face of its fluctuating environment. Most of these modules exhibit phylum-level diversity with the exception of an exclusively cyanobacterial module which is negatively correlated with the largest module, which houses all remaining bottleneck OTUs. The disposition of the cyanobacterial module suggests that the autogenic ecosystem engineers exert a limited influence upon community structure, unlike heterotrophic bottleneck OTUs. Therefore, we contend that filamentous cyanobacteria, having engineered the cryoconite ecosystem by the aggregation of aeolian organic matter and inorganic debris (Hodson *et al.* 2010; Langford *et al.* 2010) are disconnected from the heterotrophic bacterial community, which comprises closely interacting taxa. As such, the assembly of the cryoconite bacterial community is biotically filtered with a primary succession from phototrophic taxa associated with granule formation towards a highly interactive consortium of heterotrophic bacteria which may act to humify the accumulated organic matter.

### *Technical considerations and limitations of the present study*

This study focuses upon the intensive coverage of one ice cap in the High Arctic at a single time point and targets the bacterial community only. We note the presence of eukaryote and viral communities in cryoconite (Säwström *et al.* 2002), beyond the scope of this study, as well as reliable reports of Archaea associated with cryoconite, albeit from Alpine and Antarctic cryoconite (Cameron *et al.* 2012; Hamilton *et al.* 2013). The amplicon sequencing data set generated within this study did not reveal the presence of Archaea, and PCR assays targeting Archaea did not generate specific amplicons (data not shown). In line with other work suggesting that Archaea are not detected in Arctic cryoconite (Edwards *et al.* 2011; Cameron *et al.* 2012), we focused upon the bacterial community. Furthermore, analyses of multiple time points and localities are likely to yield further insights to the structure and function of glacial ecosystems (Edwards & Cook 2015). However, recent studies implying the temporal stability of bacterial communities in cryoconite ecosystems (Musilova *et al.* 2015), coupled with the cosmopolitan distribution of core taxa (Table S6, Supporting information) documented in a variety of cryospheric habitats imply the broader utility of insights from the Foxfonna ice cap. The importance of melt season duration as a physical parameter predicting the bacterial community structure is noted, and thus, examination of cryoconite ecosystems in discrete stages of melt season conditions is recommended, as this study sampled during the late melt season.

Bulk DNA extraction coupled with amplicon semiconductor sequencing of the 16S rRNA gene has been employed, in line with many other contemporary studies in microbial ecology (Prosser 2012). Necessarily these studies all entail systematic biases in extraction, amplification and sequencing (Lee *et al.* 2012). An important caveat here is that bulk DNA extracts will include templates from active and inactive taxa (Blazewicz *et al.* 2013; Klein 2015), and thus, the detection of temporal variation in activity levels is precluded (Stibal *et al.* 2015). Finally, processed reads were aligned to the GreenGenes taxonomy as described. It should be noted that the highly dominant OTU, *Leptolyngbya*-40205, assigned to the genus *Leptolyngbya* within the GreenGenes taxonomy possesses a closest named relative within an Antarctic strain *Phormidesmis priestleyi* (95% id AY493581) which is also the closest relative of Sanger-sequenced clone library Oscillatorean cyanobacterial OTUs from cryoconite elsewhere on Svalbard (Edwards *et al.* 2011) and dominates the active bacterial community of cryoconite on the southwestern margin of the Greenland Ice Sheet (Cook *et al.* 2016). As the

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phylogenetic placement of cyanobacteria from the cold biosphere improves (Chrismas *et al.* 2015), so will the taxonomic affiliation of this cryoconite ecosystem engineer, which currently resides within the *Phormidesmis*-like clade of cold adapted cyanobacteria.

### *Summary*

We conclude that the assembly of the bacterial community of microbial–mineral aggregates colonizing the High Arctic ice cap of Foxfonna is driven principally by biotic filtering. A dominant generalist core of taxa emerges which includes filamentous cyanobacterial ecosystem engineers and a discrete group of keystone taxa principally within the *Actinobacteria*, likely humifying accumulated organic matter to darken the cryoconite. While there is evidence for a moderate distance–decay effect in community similarity, it is notable that the core taxa possess close environmental relatives from the global cryosphere, linking microbial colonization processes interacting with glacier change at local scales with dispersal within the cosmopolitan cold biosphere (Jungblut *et al.* 2009).

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A.J.H., J.K.G. and A.E. conceived study. A.E., T.I.F., J.S.B., N.T. and A.J.H. conducted fieldwork. A.E., J.K.G., P.J.W. and A.P.D. conducted laboratory work. J.K.G., E.R.S., T.I.F., L.A.J.M. and A.E. conducted data analyses. J.K.G., E.R.S., T.I.F. and A.E. wrote the manuscript. All authors contributed to and approved the manuscript submission.

## Data accessibility

Sequence data are available at EBI-SRA (SRP067436: PRJNA306097).

## Supporting information

Additional supporting information may be found in the online version of this article.

**Fig. S1** Distance decay relationship between pairwise Bray Curtis distance between 16S sequencing profiles and pairwise physical distance.

**Table S1** Modified primers used for amplification of bacterial 16S rRNA gene regions (5′–3′) V1–V3.

**Table S2** Diversity indices and environmental parameters measured.

**Table S3** Pairwise PERMANOVA comparison of bacterial community structure (at the OTU level) by sector of Foxfonna ice cap.

**Table S4** Marginal tests from distLM model of environmental predictors of bacterial community structure.

**Table S5** Sequential tests from distLM model of environmental predictors of bacterial community structure. SS: sums of squares; Pseudo-F: test statistic.

**Table S6** BLAST matches of core OTUs to closest environmental relatives (CER) and closest named relatives (CNR).

**Table S7** Marginal tests from distLM model of core predictors of tail population structure.

**Table S8** Sequential tests from distLM model of core predictors of tail population structure.

**Table S9** Marginal tests from distLM model of environmental and core OTU predictors of total population structure.

**Table S10** Sequential tests of environmental and core OTU predictors of total community structure.

**Table S11** Sequential tests of environmental and core OTU predictors of tail community structure.

**Table S12** Marginal tests from distLM model of environmental and core OTU predictors of tail population structure.

**Data S1**. DNA extraction

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